

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation is maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files are organized into dataset, model, application, and utility folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as credit_score, income, prediction_result clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like train_model(), preprocess_data(), and predict_approval() are descriptive.

5	Code Comments	Inline and block comments explain logic	Yes	Important logic and machine learning workflow are documented using comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Data preprocessing, model training, prediction, and UI are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code has been minimized through reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Invalid inputs, missing values, and runtime exceptions are handled appropriately.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas)	Model Training & Prediction	High

2	Machine Learning Model	Scikit-learn	Prediction System	High
3	Feature Engineering Module	Python	Training & Testing	High
4	Prediction Module	Flask + Python	Web Application	High
5	User Interface	HTML, CSS, Flask	User Dashboard	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure with separate modules.
Readability	4.7	Clear naming conventions and easy-to-understand code.

Reusability	5	Functions and modules are reusable across the application.
Documentation / Comments	4	Adequate comments explaining major functions and workflow.
Overall Score	4.8	Clean, modular, maintainable, and reusable codebase.